

Continuing with the previous example, if we define a 2 x 2 matrix as

$$P \equiv (x^1 \ x^2) = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

Then $P'P = I$, i.e., P is an orthogonal matrix.

$$\begin{aligned} \text{and } P'AP &= \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \\ &= \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \Lambda \end{aligned}$$

i.e., $P'AP = \Lambda$, where Λ is the diagonal matrix consisting of the characteristic values of A as its diagonal elements.

7.4 LINEAR INDEPENDENCE OF EIGEN VECTORS

Another important property, in this context, is that if $\lambda_1, \lambda_2, \dots, \lambda_n$, are distinct, then the corresponding characteristic vectors viz. $x_1, x_2, \dots, x_n$, of A , are *linearly independent*. The condition (for the roots to be distinct) is sufficient but not necessary. Even when some root, or more than one root, is repeated, it may be possible to find a set of n characteristic vectors, which are linearly independent.

Moreover, if two roots are distinct then the characteristic vectors corresponding to them will be orthogonal.

7.5 QUADRATIC FORMS

Consider the expression $a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2$ where a_{11} , a_{12} and a_{22} are given constants and x_1 and x_2 are real variables. Expression such as this—being homogeneous of degree 2 in x_1 and x_2 —is called **quadratic form**.

Generally, on evaluation of the expression, its sign would depend not only on a 's but also on the values of x_1 and x_2 . But sometimes, given the values of a 's, it is possible to determine the sign of the expression, whatever x_1 and x_2 may be.

For example, for $Q(x_1, x_2) = x_1^2 - 4x_1x_2 + 4x_2^2 = (x_1 - 2x_2)^2$,

the sign is positive for all values of x_1 and x_2 .

Similarly, for $Q(x_1, x_2) = -10x_1^2 + 6x_1x_2 - x_2^2 = -x_1^2 - (3x_1 - x_2)^2$,

the sign is negative for all values of (x_1, x_2) .

The **quadratic forms** that have a definite sign for all values of (x_1, x_2) are said to **be either positive definite or negative definite depending on the sign**.

Other quadratic equations, which could vary in sign, are said to be **indefinite** quadratic forms.

In general, with variables $x_1, x_2, \dots, x_n$, we may have a quadratic form as:

$$Q(x_1, x_2, \dots, x_n) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$$

Thus, $Q(\cdot)$ is **positive definite quadratic form** if $Q(x) > 0 \forall x \in \mathbf{R}_n$

Or, **positive semi-definite** if $Q(x) \geq 0 \forall x \in \mathbf{R}_n$

Identical definitions hold for **negative definite and negative semi-definite quadratic forms**.

Consider the equation: $Q(x_1, x_2) = a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2$

$$\begin{aligned} &= a_{11} \left[\left(x_1 + \frac{a_{12}}{a_{11}} x_2 \right)^2 + \left(\frac{a_{22}}{a_{11}} - \frac{a_{12}^2}{a_{11}^2} \right) x_2^2 \right] \\ &= a_{11} \left(x_1 + \frac{a_{12}}{a_{11}} x_2 \right)^2 + \left(\frac{a_{11}a_{22} - a_{12}^2}{a_{11}} \right) x_2^2 \end{aligned}$$

Clearly, $Q(x_1, x_2) > 0$ for all (x_1, x_2) if $a_{11} > 0$ and $\frac{a_{11}a_{22} - a_{12}^2}{a_{11}} > 0$

Thus, $a_{11} > 0$, $a_{11}a_{22} - a_{12}^2 > 0$ are sufficient conditions for the quadratic form Q to be positive definite.

These are *necessary* conditions also, as for $a_{11} < 0$, values of Q can be less than zero with high value of x_1 and low value of x_2 , both in magnitudes. Also, for $(a_{11}a_{22} - a_{12}^2) < 0$, Q can be less than zero if $a_{11} < 0$.

7.6 DEFINITENESS AND EIGEN VALUES

A necessary and sufficient condition for

Q to be *positive semi-definite* is that $|A| = 0$ and $a_{11} \geq 0$, where $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$.

In terms of characteristic roots, the necessary and sufficient conditions for Q to be positive semi-definite are that $[A - \lambda I] = 0$, $\lambda_1 \geq 0$, and $\lambda_2 \geq 0$.

For Q to be negative semi definite, the corresponding conditions are $a_{11} \leq 0$ and $|A| = 0$

and in terms of characteristic roots, $\lambda_1 \leq 0$, and $\lambda_2 \leq 0$.

7.7 VECTOR DIFFERENTIATION

In some maximisation and minimisation problems, expressions involving vectors and matrices must be differentiated. The derivatives of simple expressions involving vectors and matrices can be obtained directly from the definitions. The vector derivatives of linear functions, quadratic forms and bilinear forms are defined and illustrated in this section. Second and higher order derivatives are not

discussed explicitly, but can be obtained by successive differentiation in the usual manner.

7.7.1 Vector Differentiation of a Linear Function

An expression of the form $\mathbf{a}'\mathbf{x}$, where $\mathbf{a}$ is $n \times 1$ matrix and $\mathbf{x}$ is $n \times 1$ vector, is called

a linear function. For the linear function $\mathbf{a}'\mathbf{x} = [a_1 \quad a_2 \quad \cdots \quad a_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

$$= a_1x_1 + a_2x_2 + \cdots + a_nx_n,$$

the partial derivative of $\mathbf{a}'\mathbf{x}$ with respect to the scalars x_i , $i = 1, 2, \dots, n$, are

$$\partial(\mathbf{a}'\mathbf{x})/\partial x_i = a_i,$$

that is, the partial derivatives are the elements of the vector $\mathbf{a}$. Thus, if the n partial derivatives are arranged as a vector $\mathbf{a}$, the vector differentiation is given by $\frac{\partial(\mathbf{a}'\mathbf{x})}{\partial \mathbf{x}} = \mathbf{a}$, where $\frac{\partial(\mathbf{a}'\mathbf{x})}{\partial \mathbf{x}}$ denotes the operation of differentiating $\mathbf{a}'\mathbf{x}$ with respect to the elements of the vector $\mathbf{x}$.

Check Your Progress 1

- 1) Define the terms: eigen value, eigen vector and characteristic equations.

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- 2) If $\mathbf{a} = \begin{bmatrix} 2a \\ -a \\ 3a \\ a \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ then prove that $\frac{\partial}{\partial \mathbf{x}}(\mathbf{a}'\mathbf{x}) = \mathbf{a}$

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7.7.2 Vector Differentiation of a Vector of Functions

Let $\mathbf{y}$ be an n dimensional column vector each element of which is a function of the m elements of the vector $\mathbf{x} = (x_1, x_2, \dots, x_m)$, that is, $y_i = f_i(x_1, x_2, \dots, x_m)$, $i=1, 2, \dots, n$.

Then each y_i can be differentiated partially with respect to each x_j and these partial derivatives can be arranged in an $m \times n$ matrix as follows:

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_2}{\partial x_1} & \dots & \frac{\partial y_n}{\partial x_1} \\ \frac{\partial y_1}{\partial x_2} & \frac{\partial y_2}{\partial x_2} & \dots & \frac{\partial y_n}{\partial x_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_1}{\partial x_m} & \frac{\partial y_2}{\partial x_m} & \dots & \frac{\partial y_n}{\partial x_m} \end{bmatrix}$$

Example: If $\mathbf{y} = \begin{bmatrix} x_1^2 + 2x_2^2 - 3x_1x_3 \\ 3x_1x_2^2 - x_2^2 + 4x_2x_3^2 \end{bmatrix}$,

find vector differentiation of $\mathbf{y}$ w.r.t the vector $\mathbf{x} = [x_1, x_2, x_3]$.

Here, the vector $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$,

where $y_1 = x_1^2 + 2x_2^2 - 3x_1x_3$, and

$$y_2 = 3x_1x_2^2 - x_2^2 + 4x_2x_3^2.$$

$$\text{Then } \frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \begin{bmatrix} 2x_1 - 3x_3 & 3x_2^2 \\ 4x_2 & 6x_1x_2 - 2x_2 + 4x_3^2 \\ -3x_1 & 8x_2x_3 \end{bmatrix}$$

7.7.3 Vector Differentiation of a Quadratic Form

An expression of the form $\mathbf{x}'\mathbf{A}\mathbf{x}$, where $\mathbf{a}$ is an $n \times n$ symmetric matrix, is a quadratic form. The quadratic form $\mathbf{x}'\mathbf{A}\mathbf{x}$ can be expanded as follows

$$\mathbf{x}'\mathbf{A}\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{12} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$= a_{11}x_1^2 + a_{12}x_1x_2 + 2a_{13}x_1x_3 + \dots + 2a_{1n}x_1x_n + a_{22}x_2^2 + 2a_{23}x_2x_3 + \dots + 2a_{2n}x_2x_n + \dots + a_{nn}x_n^2.$$

Taking partial derivatives with respect to the elements of $\mathbf{x}$

$$\frac{\partial}{\partial x_1}(\mathbf{x}'\mathbf{A}\mathbf{x}) = 2(a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n)$$

$$\frac{\partial}{\partial x_2}(x'Ax) = 2(a_{12}x_1 + a_{22}x_2 + \dots + a_{n2}x_n)$$

.....

$$\frac{\partial}{\partial x_n}(x'Ax) = 2(a_{1n}x_1 + a_{2n}x_2 + \dots + a_{nn}x_n)$$

Apart from the factor 2, the right-hand side of the set of equations contains the elements of Ax , which is an n dimensional column vector. Alternatively, the right-hand side contains the elements of $x'A$, which is an n dimensional row vector. Thus, $\frac{\partial}{\partial x}(X'AX) = 2AX$ and also, $\frac{\partial}{\partial x}(X'AX) = 2X'A$.

In practice, the choice between these two forms usually depends on the context in which differentiation occurs, since matrices can be equated only if they are of the same order and, specifically, a row vector cannot be set equal to a column vector.

7.7.4 Vector Differentiation of a Bi-linear Form

An expression of the form $x'Bz$, where x' is $1 \times m$, B is $m \times n$, and z is $n \times 1$, is a bilinear form. The following derivatives for a bilinear form can be verified using the procedures above.

$$\frac{\partial}{\partial x}(x'Bz) = Bz \text{ and } \frac{\partial}{\partial z}(x'Bz) = B'x.$$

Check Your Progress 2

1) If $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $A = \begin{bmatrix} 3 & 1 & -2 \\ 1 & 0 & 3 \\ -2 & 3 & 2 \end{bmatrix}$ then show that $\frac{\partial}{\partial x}(x'Ax) = 2x'A$

.....

2) If $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $B = \begin{bmatrix} 2b_1 & b_1 + b_2 \\ b_2 + 2b_3 & 3b_2 \\ b_1 + b_3 & 4b_3 \end{bmatrix}$ $z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$

then $\frac{\partial}{\partial x}(x'Bz) = Bz$ and $\frac{\partial}{\partial z}(x'Bz) = B'x$

.....

7.8 LET US SUM UP

In this unit, first we discussed how to present a system of linear equations as a matrix equation. Then, we discussed the concepts of characteristic matrix, characteristic vector; characteristic root, eigen vector, eigen value etc. and explained the characteristic value problem. Next, we discussed the concept of quadratic form; and briefly discussed its various types. Next, we discussed the relation of eigen values to definiteness of a quadratic form. Further, we discussed the concept of vector differentiation, and explained the processes of various types of vector differentiation, including that of a linear function, quadratic forms and bilinear forms.

7.9 KEY WORDS

Characteristic Equation

: For given n , to solve the system of equations $Ax = \lambda x$, for λ , it can be written as $(A - \lambda I)x = 0$, where λI is a unit matrix of order n . Now, if the system has to have a non-trivial solution for the variable vector X , then $\text{rank}[A - \lambda I] < n$, which is true if and only if

$$\begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{vmatrix} = 0$$

This gives us an n^{th} degree polynomial equation in λ . This is called the '**characteristic equation**' of A and the left-hand side is called the '**characteristic polynomial**'.

Eigen Value and Eigen Vector

: If there exist a vector $X \neq 0$ (i.e. non-null vector) satisfying $Ax = \lambda x$, where A is a given n^{th} order square matrix and λ is a scalar parameter, then such a problem is called a eigen value problem and x is called the eigen vector associated with A if $Ax = \lambda x$.

Quadratic Forms

: Consider the expression $a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2$ where a_{11} , a_{12} and a_{22} are given constants and x_1 and x_2 are real variables. Expression such as this--being homogeneous of degree 2 in x_1 and x_2 —is called

7.10 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) See the text and answer
 2) [Hint: $\mathbf{a}'\mathbf{x} = 2ax_1 - ax_2 + 3ax_3 + ax_4$, then find

$$\frac{\partial}{\partial x_1}(\mathbf{a}'\mathbf{x}) = 2a, \quad \frac{\partial}{\partial x_2}(\mathbf{a}'\mathbf{x}) = -a, \quad \frac{\partial}{\partial x_3}(\mathbf{a}'\mathbf{x}) = 3a \text{ and } \frac{\partial}{\partial x_4}(\mathbf{a}'\mathbf{x}) = a]$$

Check Your Progress 2

- 1) [Hint: $\mathbf{x}'\mathbf{A}\mathbf{x} = 3x_1^2 + 2x_3^2 + 2x_1x_2 - 4x_1x_3 + 6x_2x_3$ then find

$$\frac{\partial}{\partial x_1}(\mathbf{x}'\mathbf{A}\mathbf{x}) = 6x_1 + 2x_2 - 4x_3, \quad \frac{\partial}{\partial x_2}(\mathbf{x}'\mathbf{A}\mathbf{x}) = 2x_1 + 6x_3 \text{ and}$$

$$\frac{\partial}{\partial x_3}(\mathbf{x}'\mathbf{A}\mathbf{x}) = -4x_1 + 6x_2 + 4x_3]$$

- 2) Follow the same logic as given above.

7.11 EXERCISES

Q1. Derive the norm of the following vectors if

(i) $\mathbf{a} = (0 \ 2 \ 1)$; $\mathbf{b} = (6 \ 3 \ 2)$; $\mathbf{c} = (-5 \ 4 \ 3)$; $\mathbf{d} = (0 \ 0 \ 1)$; $\mathbf{e} = (-3 \ -4 \ -2)$.

Ans: (i) 5; (ii) 49; (iii) 50; (iv) 1; (v) 29.

Q2. If $\mathbf{b} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, find $\frac{\partial}{\partial \mathbf{x}}(\mathbf{b}'\mathbf{x})$.

Ans: $\begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$

Q3. If $\mathbf{c} = \begin{bmatrix} c^2 + 1 \\ 3c \\ 4c - 5 \\ c^3 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$, find $\frac{\partial}{\partial \mathbf{x}}(\mathbf{c}'\mathbf{x})$.

Ans: $\begin{bmatrix} c^2 + 1 \\ 3c \\ 4c - 5 \\ c^3 \end{bmatrix}$

Q4. If $y = \begin{bmatrix} x_1^2 + 3x_2 \\ 2x_1x_2 - x_2 \\ 2x_1 + x_1x_2 - 3x_2^2 \\ 3x_1^2 - x_1^2x_2 - x_2^3 \end{bmatrix}$, find $\frac{\partial y}{\partial x}$.

Ans: $\begin{bmatrix} 2x_1 & 2x_2 & 2 + x_2 & 6x_1 - 2x_1x_2 \\ 3 & 2x_1 - 1 & x_1 - 6x_2 & -x_1 - 3x_2^2 \end{bmatrix}$

Q5. If $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, $A = \begin{bmatrix} 3 & 0 & -1 \\ 2 & 1 & 4 \\ -1 & 0 & 3 \end{bmatrix}$ and $y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$,

find $\frac{\partial}{\partial x}(x' Ay)$ and $\frac{\partial}{\partial y}(x' Ay)$.

Ans: $y = \begin{bmatrix} 3y_1 - y_3 \\ 2y_1 + y_2 + 4y_3 \\ -y_1 + 3y_3 \end{bmatrix}$

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UNIT 8 VECTOR AND MATRIX REPRESENTATIONS OF LINEAR EQUATIONS

Structure

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- 8.2 System of Linear Equations
 - 8.2.1 Linear Equation
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8.0 OBJECTIVES

After going through this unit, you should be able to

- explain the concept of a system of linear equations and its matrix representation;
- discuss elementary row operations of a matrix;
- discuss and test for linear independence of m equations in n variables;
- explain and test the concept of consistency of a system of linear equations; and
- use the concept of rank of a matrix for testing a given system of equations for linear independence and consistency.

8.1 INTRODUCTION

Systems of linear equations play a significant role in solving problems in a number of academic disciplines including engineering, physics, chemistry, computer science, specially, in computer graphics, genomics, and, of course, in economics. For example, the relation of supply and demand may be considered, in most simplified form, as a linear equation:

$$\text{Demand} = a - b \text{ Price} + c \text{ Income},$$

in which a , b , c are constants, and Demand, Price, and Income are variables.

Another example, where system of linear equations may be useful: Assume that a firm can sell all its products it manufactures in a month at Rs. 30, 000 each. It has to pay out Rs. 5 lacs as fixed costs. Also, the firm must pay a marginal cost of Rs. 5, 000 for each unit produced. How much does the firm need produce to break-even?

Even those complex systems, which can be represented only by a system of non-linear equations, can also be approximated by a system of linear equations.

In this respect, a number of questions arise: What is a system of equations? Does every system of linear equations have a solution? If a solution exists, is the solution unique?

This unit—using the concepts of matrix and vector, as tools, for representation of a system of linear equations--intends to answer these questions, apart from describing processes of obtaining solutions under given constraints.

8.2 SYSTEM OF LINEAR EQUATIONS

We start our discussion with some definitions.

8.2.1 Linear Equation

An equation, in which each of the occurring variables has degree not exceeding one, at least one variable has degree one, and coefficient of at least one variable with degree one, is non-zero. Each of the following is a linear equation

- $7x = 4$, is a linear equation involving one variable x and two constant real numbers 7 and 4;
- $a_1 x_1 = b_1$, is a linear equation in which x_1 is a variable, and a_1 , and b_1 are parameters;
- $a_1 x_1 - 5 x_2 = 11$, is a linear equation involving two variables x_1, x_2 ; a parameter a_1 , and two constants -5 and 11 .

*Such equations are called **linear**, because each of the above equation represents a straight **line** in 2-dimensional co-ordinate plane.*

- In general, for a given positive integer n , a linear equation is of the form

$$a_1 x_1 + a_2 x_2 + \dots + a_n x_n = b,$$